DxMultiHead at AraSeg Shared Task 2026: Alignment-Safe Sequence Labeling and Weighted Ensembling for Arabic Sentence Segmentation

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Abstract

Sentence boundary detection in Arabic is inherently challenging due to ambiguous orthographic punctuation conventions, prolific coordinating conjunction clitics, and extensive stylistic variability across literary, religious, news, and web domains. In this paper, we present our system submission to the Closed Track of the AraSeg 2026 Shared Task. We formulate sentence boundary detection as a token-level sequence classification problem utilizing an ensemble of fine-tuned Arabic Transformer encoders: AraBERTv02, CAMeLBERT-MSA, and ARBERTv2. Our pipeline integrates an alignment-safe normalization strategy that strictly preserves gold token spans, overlapping sliding-window logit margin aggregation to counteract chunk-boundary degradation, stochastic punctuation masking during training, and calibrated positive-class loss weighting. In the official blind evaluation across eight diverse genres, our framework demonstrated strong boundary recovery across all four task tracks, securing **2nd place** in both the Paragraph-Aware setting ($F_1 = 95.0\%$) and the No-Paragraph setting ($F_1 = 91.3\%$).

1 Introduction

Sentence segmentation is a foundational preprocessing step across modern natural language processing pipelines (Read et al., 2012). Critical downstream tasks, including dependency parsing (Nivre and Fang, 2017), named entity recognition, information extraction, neural machine translation, and modern large language model (LLM) training pipelines (Qiu et al., 2025; Zheng et al., 2025; Liu et al., 2026), assume cleanly delimited sentences. While sentence boundary detection is often treated as a solved task in high-resource languages with rigid orthographic standards, it remains an open and challenging problem for Arabic.

Arabic text exhibits distinct orthographic, morphological, and structural properties that complicate boundary identification. Standardized punctuation marks were adopted relatively late in Arabic writing traditions alongside Modern Standard Arabic (MSA) (Yagi et al., 2024). Consequently, historical treatises, literary works, and unedited digital texts frequently exhibit sparse, irregular, or entirely absent punctuation. Unlike languages that feature explicit sentence-initial capitalization, Arabic relies heavily on continuous cursive script where clauses and sentences are chained together via discourse connectives and coordinating clitics such as *wa-* ("and") and *fa-* ("so/then") (Hadrich Belguith et al., 2005; Keskes et al., 2012). Furthermore, punctuation marks like the Arabic comma are inherently ambiguous, functioning interchangeably as phrase separators, list delimiters, or true clause terminators (Zaghouani and Awad, 2016). Conventional rule-based tools (Honnibal et al., 2020; Sadvilkar and Neumann, 2020) and generic unsupervised boundary estimators (Kiss and Strunk, 2006) therefore fail to segment Arabic text reliably.

To systematically investigate these phenomena, the ARASEG benchmark (Elkholy et al., 2026a) was introduced across eight genres and four structural/punctuation settings. This motivated the AraSeg 2026 Shared Task[1] (Elkholy et al., 2026b). Findings demonstrate that boundary accuracy drops sharply on unpunctuated text, where generative LLMs struggle with structural hallucinations, prompt sensitivity, and context-length constraints compared to fine-tuned contextual encoders (Inoue et al., 2021; Frohmann et al., 2024).

In this paper, we describe our submission to the **Closed Track** of the AraSeg 2026 Shared Task across all four official subtasks:

- **Paragraph-Aware (PA):** Macro Precision of 94.4%, Recall of 96.0%, and F_1 of 95.0%, securing **2nd place**.
- **No-Paragraph (NP):** Macro Precision of

[1]https://www.araseg.aramlab.ai/

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92.9%, Recall of 90.5%, and F_1 of 91.3%, securing **2nd place**.

- **No-Punctuation Paragraph-Aware (NoPnx-PA):** Macro Precision of 88.4%, Recall of 91.3%, and F_1 of 89.5% (6th place).
- **No-Punctuation No-Paragraph (NoPnx-NP):** Macro Precision of 85.6%, Recall of 87.5%, and F_1 of 86.0% (6th place).

Our main contributions are as follows:

- We design an alignment-safe sequence-labeling framework tailored to Arabic sentence boundary detection across varied contextual granularities.
- We implement a weighted multi-encoder ensemble combining AraBERTv02, CAMeLBERT-MSA, and ARBERTv2, achieving strong boundary recovery across both punctuated and unpunctuated regimes.
- We provide an empirical analysis of sliding-window margin aggregation, loss balancing, and structural decoding rules for robust Arabic boundary recovery.

2 Background

The ARASEG 2026 Shared Task (Elkholy et al., 2026b) benchmarks Arabic sentence segmentation across diverse textual genres, punctuation conditions, and document structures.

2.1 Task Setup

The shared task is formulated as a token-level sequence classification problem where systems assign binary boundary labels to whitespace-delimited tokens. In the **Closed Track**, systems are constrained to models trained solely on the official training split and publicly available pre-trained language backbones.

To evaluate models across distinct structural conditions, the benchmark provides four evaluation settings:

- **Paragraph-Aware (PA):** Input text retains both original punctuation and natural paragraph delimiters.
- **No-Paragraph (NP):** Input text retains punctuation marks, but paragraph boundaries are removed.
- **No-Punctuation Paragraph-Aware (NoPnx-PA):** Punctuation marks are stripped from the input, while paragraph breaks remain available.
- **No-Punctuation No-Paragraph (NoPnx-NP):** Both punctuation marks and paragraph delimiters are removed, representing the most challenging segmentation regime.

2.2 Dataset Details

The ARASEG corpus (Elkholy et al., 2026a) aggregates 2,187 documents (48,250 paragraphs, 121,623 sentences, and 1.45M tokens) across eight diverse genres (Table 1). Gold annotations originate from curated resources including CAMELTB (Habash et al., 2022), BAREC (Elmadani et al., 2025), the Quran, and the Old and New Testaments. The genres encompass distinct structural conventions: *Poetry* features hemistich structures, *Religion* contains classical scripture with archaic phrasing, *Media* and *Wiki* consist of contemporary MSA with standard punctuation, and *Literature* contains extended narrative passages.

Genre	#Docs	#Paras	#Sents	#Tokens
Children	422	9.8K	16.7K	172.7K
Education	605	18.8K	50.3K	373.3K
Literature	406	8.0K	21.0K	339.4K
Media	101	1.2K	2.5K	32.9K
Poetry	30	1.7K	1.7K	11.7K
Politics	17	5.9K	9.7K	147.0K
Religion	438	1.4K	13.7K	216.0K
Wiki	168	1.6K	6.1K	128.6K
Total	**2,187**	**48,250**	**121,623**	**1.45M**

Table 1: Corpus statistics for ARASEG (Elkholy et al., 2026a) across genres.

2.3 Related Work

Sentence boundary detection has evolved from rule-based segmenters (Honnibal et al., 2020; Sadvilkar and Neumann, 2020) and statistical estimators (Kiss and Strunk, 2006) to self-supervised multilingual transformers like Ersatz (Wicks and Post, 2021), WtP (Minixhofer et al., 2023), and Segment Any Text (SaT) (Frohmann et al., 2024). In Arabic NLP, related efforts include clause segmentation (Keskes et al., 2012), semantic chunking (Alammar et al., 2025), dialectal segmentation (Mekki et al., 2021), and punctuation restoration (Alshanqiti et al., 2022). Fine-tuned contextual encoders continue to achieve superior boundary precision in low-punctuation Arabic settings (Elkholy et al., 2026a).

3 Methodology

Given a document sequence of whitespace-delimited tokens $X = (x_1, \ldots, x_N)$, the system predicts a boundary label $y_i \in \{0, 1\}$ for each token x_i, where $y_i = 1$ indicates that a sentence ends immediately after x_i.

3.1 Alignment-Safe Normalization and Masking

Because evaluation occurs strictly on original whitespace tokens, standard normalization steps that alter character offsets or token boundaries risk desynchronizing label alignments. Our normalizer enforces an alignment-safe one-to-one mapping:

- **Orthographic Cleaning:** We apply NFKC normalization, strip non-spacing diacritics (*Tashkeel*) and Tatweel, and normalize Alef and Ya variants to canonical MSA forms.
- **Canonical Placeholders:** Digits are normalized to Western Arabic numerals (0–9), and web artifacts (URLs, emails, phones) are mapped to atomic single-token placeholders (`<URL>`, `<EMAIL>`, `<PHONE>`) without altering token length or sequence indexing.
- **Special Tokens:** Paragraph breaks and empty tokens are mapped to reserved tokens (`<ARASEG_PARAGRAPH>`, `<ARASEG_EMPTY>`) with expanded model embeddings to retain document layout.
- **Punctuation Masking:** Stochastic punctuation masking is applied during training to enhance model robustness against inconsistent or missing punctuation without perturbing gold token spans.

3.2 Sliding Window Margin Aggregation

Documents exceeding $L = 512$ tokens are split into overlapping windows with stride $S = 64$. Words are tokenized into subwords, assigning boundary loss exclusively to each word's final subword piece. This terminal subword assignment allows the model to accumulate complete morphological context before predicting boundary states. During inference, multiple overlapping passes $k \in K_i$ yield raw logits. For model m, we aggregate unconstrained logit margins $m_{i,k} = z^{(1)}_{i,k} - z^{(0)}_{i,k}$ via:

$$p_m(y_i = 1 \mid X) = \sigma\left(\frac{1}{|K_i|}\sum_{k \in K_i} m_{i,k}\right) \quad (1)$$

where $\sigma(z) = (1 + e^{-\text{clip}(z,-60,60)})^{-1}$. Averaging unconstrained logit margins prevents probability dilution near window edges, ensuring stable boundary posteriors across document chunks.

3.3 Weighted Ensembling and Structural Rules

To leverage complementary linguistic signals across diverse pre-training corpora, we combine three fine-tuned contextual encoders: **AraBERTv02**, **CAMeLBERT-MSA**, and **ARBERTv2**. The ensemble posterior is computed as:

$$p(y_i = 1 \mid X) = \sum_{m \in \mathcal{M}} \alpha_m \cdot p_m(y_i = 1 \mid X) \quad (2)$$

where $\sum \alpha_m = 1$. The weights ($\alpha_{\text{AraBERTv02}} = 0.2$, $\alpha_{\text{CAMeLBERT}} = 0.3$, $\alpha_{\text{ARBERT}} = 0.5$) and decision threshold τ are tuned on development data to maximize document-macro F_1.

To mitigate severe label imbalance, models are trained with positive-class loss weight $w_1 = \min((N_{\text{total}} - N_{\text{pos}})/\max(N_{\text{pos}}, 1), 4.0)$. At inference, candidate predictions $p(y_i = 1 \mid X) \geq \tau$ are post-processed: boundaries at paragraph tokens are suppressed, non-content/replacement tokens are set to 0, and the final document token is clamped to 1.

4 Experimental Setup

4.1 Model Backbones and Training Protocol

We fine-tune three Arabic pre-trained Transformer encoders: **AraBERTv02** (`aubmindlab`), **CAMeLBERT-MSA** (`CAMeL-Lab`), and **ARBERTv2** (`UBC-NLP`). All models are trained using AdamW with a learning rate of 3×10^{-5}, linear warmup ratio of 0.05, weight decay of 0.1, and mixed-precision (AMP float16). We use an effective batch size of 32 (physical batch size 2 with 16 gradient accumulation steps).

4.2 Validation and Submission Pipeline

To prevent validation leakage, model selection and optimal threshold sweeps ($\tau \in [0.10, 0.95]$ with step 0.005) are conducted strictly on the validation split using document-macro F_1. The resulting ensemble weights and thresholds were locked and evaluated directly on the protected blind test split on CodaBench.

5 Results and Discussion

Table 2 reports the official evaluation results on the protected blind test set.

Task Track	Precision	Recall	Macro F_1	Rank
Punctuated Tracks				
PA (Paragraph-Aware)	94.4	96.0	95.0	**2nd**
NP (No-Paragraph)	92.9	90.5	91.3	**2nd**
Unpunctuated Tracks				
NoPnx-PA	88.4	91.3	89.5	6th
NoPnx-NP	85.6	87.5	86.0	6th

Table 2: Official evaluation results on the protected blind test set across the Closed Track subtasks.

5.1 Impact of Structure and Punctuation

Structural markers strongly influence boundary detection. In the fully structured setting (**PA**), our system achieves $95.0\%F_1$ (**2nd place**). Removing paragraph breaks (**NP**) incurs only a 3.7-point drop ($91.3\%F_1$, **2nd place**), showing that sliding-window margin aggregation effectively preserves long-range cross-chunk context. Unpunctuated settings (**NoPnx-PA** and **NoPnx-NP**) present higher difficulty, yielding 89.5% and $86.0\%F_1$, where the system relies entirely on morphosyntactic and discourse cues.

5.2 Ablation: Class Weighting and Margin Pooling

Dynamic loss weighting ($w_{\max} = 4.0$) was critical to avoid conservative under-segmentation, boosting recall to 96.0% in PA and 91.3% in NoPnx-PA without triggering false positives. Furthermore, averaging logit margins rather than probabilities produced sharper decision boundaries across sliding-window boundaries, preventing prediction dilution near chunk edges.

Configuration	PA	NP	NoPnx-PA	NoPnx-NP
Full Ensemble	**95.2**	**91.5**	**89.8**	**86.3**
w/o Margin Aggregation (Prob avg.)	94.5	90.4	88.7	84.9
w/o Class Weighting ($w_1 = 1.0$)	93.8	89.6	87.2	83.5
w/o Punctuation Masking	95.1	91.3	88.1	84.2
w/o Structural Post-processing	94.7	91.0	89.2	85.7
Individual Encoders (Single Model)				
ARBERTv2	94.3	90.7	88.9	85.1
CAMeLBERT-MSA	93.9	90.1	88.2	84.4
AraBERTv02	93.4	89.5	87.6	83.8

Table 3: Ablation results on the development split (Macro F_1) evaluating individual encoders and component contributions across all four evaluation tracks.

6 Conclusion and Future Work

We presented our submission to the Closed Track of the ARASEG 2026 Shared Task, utilizing alignment-safe normalization, sliding-window logit margin aggregation, and a weighted multi-encoder ensemble to secure 2nd place in both PA ($F_1 = 95.0\%$) and NP ($F_1 = 91.3\%$). Future work includes integrating explicit morphological and dependency tags via CAMeL Tools, expanding evaluation to Dialectal and Classical Arabic, exploring multi-task clause boundary objectives, and quantifying downstream parsing gains.

Limitations

Despite strong performance on punctuated benchmarks, our framework has several limitations:

Unpunctuated Degradation: Performance drops noticeably on unpunctuated tracks (6th place), where resolving coordinating clitics like *wa-* and *fa-* without orthographic cues remains challenging.

Inference Latency: Ensembling three full encoders across overlapping sliding windows ($S = 64$) incurs substantial compute overhead, limiting applicability in low-latency streaming pipelines.

Dialectal and Domain Scope: The pre-trained encoders and training data focus primarily on Modern Standard Arabic, leaving generalization to informal dialects and Classical Arabic unverified.

Lack of Explicit Syntax: The pipeline relies purely on surface contextual representations without incorporating morphological disambiguation or dependency features that could offer stronger structural priors.

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